

HITHA SHAMASUNDAR

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EDUCATION

Master of Science in Software Engineering

Arizona State University (GPA: 3.93/4.0)

May 2026

Tempe, AZ

Bachelor of Engineering in Computer Science and Engineering

NIE Institute of Technology - Visvesvaraya Technological University (CGPA: 8.18/10.0)

August 2023

Bengaluru, India

SKILLS

Programming Languages: Python, Java, JavaScript, C++

Methodology: Agile, SDLC, STLC, Waterfall

Web Technologies: HTML, CSS, TypeScript, Node JS, Next JS, Nest JS React JS, JSON, Bootstrap, Flask, Vue, Kotlin

API Development: REST API, GraphQL, Nest JS

Database: MySQL, PostgreSQL, NoSQL

DevOps & Cloud: Docker, Git, Jenkins, CI/CD, GitHub, GitLab, Kubernetes, AWS

Testing tools & Automation frameworks: Postman, REST Assured, Selenium, Cucumber, Cypress, JUnit, TestNG, JMeter, Appium, Mocha & Chai

EXPERIENCE

Software Engineer (SDE/SDET),

August 2023 - June 2024

Happiest Minds Technologies Limited

Bengaluru

- Architected and developed DocBot, a AI-powered conversational application enabling natural language querying over uploaded documents using a **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain** and **OpenAI** embeddings, improving contextual retrieval relevance by 40%.
- Designed the semantic retrieval pipeline by implementing document parsing, text chunking, embedding generation, and vector indexing with FAISS, enabling efficient similarity-based vector search and improving contextual response accuracy by 35%.
- Built scalable backend services in **Python Flask** for document ingestion, vector indexing, and LLM query orchestration, exposing functionality via a lightweight frontend interface while enabling document processing and reducing query response latency by 30%.
- Engineered a modular browser automation test framework using **Selenium** integrated with **Cucumber**, implementing reusable Page Object model (POM) abstractions and behavior-driven test suites that increased regression coverage by 30%.
- Integrated **CI/CD** pipelines in **Jenkins** with automated build and test triggers via **Maven** and **Gradle**, while managed **Git**-based version control, including branching, merging, and conflict resolution across multiple repositories, which streamlined deployments.
- Collaborated in an **agile environment** to evaluate software systems for the Generative AI business unit using emerging technologies and AI-driven testing approaches & metrics, strengthening continuous quality validation.

Software Development Engineer Associate Intern

March 2023 - June 2023

HashedIn by Deloitte

Bengaluru

- Engineered a full-stack project management application using **Next.js** and **NestJS**, supporting task creation, issue tracking, story point assignment, and role-based collaboration across 50+ concurrent project items.
- Developed and maintained robust **RESTful API** endpoints in **NestJS** using a modular controller-service architecture with DTO-based validation, supporting full CRUD operations for projects, tasks, and user management - improving API performance by 50% and code maintainability by 25%.
- Built a responsive, component-driven frontend using **React** Hooks and **Next.js** routing, managing application state for task updates, issue tracking, and user interactions, reducing client-side page reloads and improving navigation responsiveness by 35%.
- Designed relational data models using **Prisma ORM** with **PostgreSQL**, implemented schema migrations and optimized queries, which improved query performance and ensured data consistency across multi-table storage for projects, tasks, users, and stories.
- Implemented secure authentication and authorization mechanisms using **JWT** and **OAuth 2.0**, enabling role-based access control (RBAC) across 3+ user permission levels and improving system security and scalability for multi-user collaboration.
- Validated and tested API endpoints using **Postman** and **Swagger**, ensuring correct request/response schemas across service endpoints and improving reliability during iterative development cycles.

PROJECTS

Dynamic Traffic Density Analysis & Green Corridor Clearance

- Aimed to benefit traffic management by optimizing traffic signal timings with a specialized feature to detect emergency vehicles and priorities its passage by granting a green signal corridor.
- Implemented utilizing concepts of **TensorFlow**, **OpenCV**, **CNN**, **YOLO & Round-robin algorithms**. Real-time traffic inputs are broken down into frames and converted to grayscale to extract vehicle count and signal time durations are incremented/decremented accordingly.

Adaptive Virtual Assistant

- Engineered an adaptive AI-powered chatbot system using **Flask** and **Gemini API**, enabling personalized, context-aware responses and improving user engagement and response relevance.
- Designed and integrated a **semantic image retrieval system** processing 100K+ Wikimedia image using **OpenCLIP** and **ChromaDB**, enabling multimodal responses and improving contextual answer accuracy by 35%.
- Developed a high-throughput **asynchronous ingestion pipeline** using SQL dump processing, recursive CTE traversal, and multi-threaded processing (aiohttp + thread offloading), eliminating API rate limits and improving data processing efficiency by 50%.